

WINTER Quarter 2025
COURSE SYLLABUS - MATH 3B
Revised as of JAN 12

INSTRUCTOR	Professor CHUCK AKEMANN Office Hours: TR 12:30-2, SH 6706 Email: akemann@ucsb.edu. No phone.	
PREREQUISITES	Math 3A with a minimum grade of C.	
TEXT	<i>Calculus, Early Transcendentals, 9th ed.</i> , by James Stewart, Cengage Learning. ISBN: 978-0-357-02229-0 (Loose-leaf, UCSB bookstore) ISBN: 978-1-337-61392-7 (Hardcover)	
GRADING	Special Homework due 3/13	10%
	Midterm I, 2/4	20%
	Midterm II, 3/4	20%
	Final (Cumulative), 3/19, 12-3 pm	50%

COURSE GOALS

- Learn fundamental definitions, concepts, and theorems in integral calculus, sequences and series. Chapters 5, 6, 7, and 11 in the text.
- Develop analytical and geometrical reasoning skills.
- Develop understanding of mathematical modeling in the physical and engineering sciences.
- Develop the ability to use correct mathematical notation to formulate precise mathematical statements.
- Learn to use online tools like Wolfram Alpha and Chat GPT to solve textbook exercises and related problems.

Course Outline

- Week 1 • Review of Integration (Chapter 5.)
- Week 2 • More Chapter 5 stuff
 • Area Between Curves (6.1)

- Week 3 • Volume: Known Cross-Sections (6.2). Volume: Disc, Washer (6.2)
• Volume Cylindrical Shells (6.3) • Work (6.4)
- Week 4 • Integration by Parts (7.1)
• Trigonometric Integrals, Basic Examples (7.2)
• Trigonometric Substitution (7.3)
- Week 5 • Test I. Covers thru section 7.1
- Week 6 • Integrals Using Partial Fractions, Basic (7.4). Strategies for Integration (7.5)
• Improper Integrals (7.6)
• Sequences (11.1)
- Week 7 • Series (11.2)
• Integral Test and Estimates of Sums (11.3)
• Comparison Tests (11.4)
- Week 8 • Alternating Series (11.5)
• Absolute Convergence, Ratio/Root Tests (11.6)
• Strategies for Testing Series (11.7)
- Week 9 • Test II, covers sections 7.2-11.7
• Power Series (11.8)
• Functions as Power Series (11.9)
• Taylor and Maclaurin Series (11.10)
- Week 10 • Applications of Taylor Polynomials
• Applications
• Review

There will be two midterms and one final exam as scheduled above. Please plan accordingly as [there will not be make-up exams](#). If you are taking a class that overlaps with ours, that is your choice but please keep in mind that we will not make any special accommodation in such situations.

Accommodations for students with disabilities will be made through the [Disabled Students Program](#) (DSP).

HOMEWORK

There are 2 kinds of homework. **Special Homework** is aimed at the course goal: “Learn to use online tools such as Wolfram Alpha and Chat GPT to solve textbook exercises and related problems.” Your TA will assign problems to be solved by online methods. You should turn in these problems on the last day of class. **Traditional homework** will consist of problems from the text that you choose yourself after reading the relevant sections in the text. See the course outline above for the section numbers. I will give you a minimal list of problems for each section. You should do more than the minimal list if time permits.

Minimal list for Chapter 5 = p. 429, 1-35, 55, 57, 59.
Minimal list for Chapter 6 = p. 479 Exercises 1-18, 25-31(a), 32.
Minimal list for 7.1 = p. 490:1-17, 44, 46, 48.

ACADEMIC INTEGRITY

You should familiarize yourself with the [UCSB Student Conduct Code](#). It is expected that all students subscribe to the ideal of academic integrity, and are willing to bear individual responsibility for their work. Any submission that fulfills an academic requirement must represent a student's original work. Any act of academic dishonesty, such as cheating, plagiarism, furnishing false information, unauthorized collaboration, or misuse of course materials will subject a person to University disciplinary action as detailed in the [UCSB Office of Student Conduct](#).

In this course, I encourage you to discuss homework questions and course content with other members of the class and the teaching assistants during the discussion sections. However, it will help you to develop your critical reasoning skills if you also spend time working independently on the homework. All instances of academic misconduct will be reported to the [UCSB Office of Student Conduct](#).

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